

From Model Explanation to Data Misinterpretation: A Cautionary Analysis of Post Hoc Explainers in Business Research

Abstract

Post hoc explainers such as SHAP and LIME are widely used in business research to interpret complex machine learning (ML) models. Although they were designed to explain a *model's* decision-making process—the mapping from $\mathbf{X}$ to $\hat{Y}$ —a growing body of work treats the explanations they produce as evidence about underlying variable relationships in the data. Based on a systematic review of 181 studies, including 56 published in leading journals, we document that this practice is widespread and examine its validity. We introduce two metrics—*direction alignment* and *strength alignment*—and use simulated data with known ground truth to assess how well SHAP and LIME recover the direction and relative importance of features in the true data-generating process. Although explanations often appear reasonable *on average*, they exhibit substantial heterogeneity, with long-tailed failures that persist even when predictive accuracy is high: high accuracy is *necessary but insufficient* for reliable explanation. We show that feature correlation and the *Rashomon effect*—many equally accurate models relying on different decision-making processes—are the key drivers of misalignment, and that *agreement in explanations* across such models provides a practical, dataset-level diagnostic of reliability. Our findings caution against using post hoc explainers to *validate* hypotheses and instead position them as exploratory tools that *generate*, rather than confirm, substantive insights.

Keywords: post hoc explanation; interpretable machine learning; SHAP; LIME; Rashomon set; research methodology.

1 Introduction

The growth in data volume and complexity has made machine learning (ML) a mainstay of business analytics, particularly for prediction (Kleinberg et al. 2015, Mullainathan and Spiess 2017). Yet prediction alone is rarely the goal. Researchers and managers also want to understand their data—in particular, *how explanatory variables ($\mathbf{X}$) relate to an outcome (Y)*: for instance, which product attributes drive customer satisfaction, how price and promotion affect demand, or which borrower characteristics raise default risk. Because most high-accuracy ML models are black boxes, *post hoc explainers* have become the standard tool for peering inside them. For example, SHAP (SHapley Additive exPlanations) (Lundberg and Lee 2017) assigns each feature a Shapley value representing its contribution to a prediction relative to a baseline, while LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al. 2016) fits a sparse local linear surrogate around an instance. Both report the direction and strength with which features influence a model's predictions.

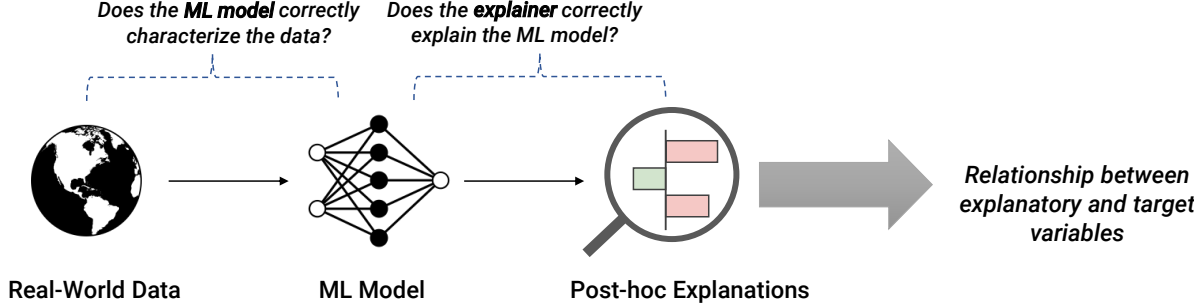


Figure 1: The explanation pipeline. A black-box model is trained for prediction and then explained by a post hoc explainer. Inferring the true marginal effects of features from these explanations—the step marked in red—constitutes a mistaken use of the pipeline.

A growing body of empirical business research adopts these explainers through a two-stage *explanation pipeline* (Figure 1): a black-box model is trained for a prediction task, and a post hoc explainer is then applied to it. Used as intended, this pipeline illuminates *model* behavior—which features the model relies on. Increasingly, however, business researchers reinterpret the same output as evidence about the *data*: they generalize explanations of the model’s mapping ($\mathbf{X} \mapsto \hat{Y}$), which are inherently model-dependent, into claims about the true relationship in the data ($\mathbf{X} \mapsto Y$). For example, [Park and Kim \(2024\)](#) read SHAP values as which smartphone features customers prefer; [Jiang et al. \(2024\)](#) treat features with large SHAP values as churn drivers; and [Sánchez-Franco and Rey-Tienda \(2023\)](#) use SHAP to identify drivers of hotel-guest satisfaction and motivate managerial action. Some work even proposes ML-plus-SHAP as an “alternative to OLS” ([Berger 2023](#)).

To gauge prevalence, we manually reviewed 181 papers that make substantive use of SHAP or LIME. Roughly **42.5%** interpret post hoc explanations as evidence of *data-level* relationships rather than statements about the model—indicating that the practice is widespread in business research; the rate is lower but still nontrivial in leading journals (16.1% in *UTD 24*, *FT50*, and *INFORMS*).

We stress that this is a *new* question, and one that is distinctively a concern for business research. Post hoc explainers have been scrutinized extensively in computer science, but that work asks whether an explanation faithfully describes the *model*—the mapping $\mathbf{X} \mapsto \hat{Y}$ it was applied to. Business research asks something that literature does not: whether the same explanation can be read as evidence about the *data*, the mapping $\mathbf{X} \mapsto Y$ in the world. The two are routinely conflated yet logically distinct, and the validity of the data-level reading—the inference our review shows empirical studies actually draw, and which drives managerial conclusions—has never been systematically tested. This motivates our central question:

To what extent can explanations derived from a machine learning model ($\mathbf{X} \mapsto \hat{Y}$) recover the direction and relative strength of the true relationship in the data ($\mathbf{X} \mapsto Y$)?

Our study proceeds in four steps. **First**, we formalize the two claims that prior business studies using post hoc explanations implicitly rely on. *Direction* interpretations assert that increasing a feature increases or decreases the outcome; *strength* interpretations rank features by importance. We give method-agnostic definitions of *direction alignment* and *strength alignment* that apply to any explainer that assigns a value to each feature. **Second**, we evaluate SHAP and LIME on simulated data where the ground-truth $\mathbf{X} \mapsto Y$ is known and precisely controlled. Although explanations look truthful on average, their alignment with the ground truth is highly heterogeneous, with pronounced long-tailed deviations—even when the black-box model’s predictive accuracy is high—so that a researcher typically cannot tell *when* an explanation is trustworthy. **Third**, we trace the drivers of

misalignment to (i) the predictive performance of the model, (ii) the *Rashomon effect* (Semenova et al. 2022, Paes et al. 2023), and (iii) data properties such as feature correlation. The central lesson is that high accuracy is *necessary but insufficient*: equally accurate models can rely on markedly different decision-making processes, thus yielding different explanations, and feature correlation is the dominant data-level driver of disagreement. **Fourth**, we show that *disagreement among equally accurate models* is a usable diagnostic. Agreement based on *explanations* is strongly correlated with alignment, whereas prediction-based agreement is not. Rashomon agreement thus provides a practical, dataset-level signal of when explanations of $\mathbf{X} \mapsto \hat{Y}$ are unlikely to recover $\mathbf{X} \mapsto Y$.

Overall, we caution against a growing trend of using post hoc explanations to draw inferences about data. Explanations should not be used to **validate** hypotheses about $\mathbf{X} \mapsto Y$; their proper role is to **propose** hypotheses that are then tested with methods built for identification—as a growing number of studies already do (Zhang and Luo 2023, Feng et al. 2025, Senoner et al. 2022). Keeping exploration and validation distinct avoids overstating the reliability of explainer-derived insights.

The rest of the paper is organized as follows. Section 2 reviews SHAP, LIME, and the Rashomon effect. Section 3 defines the data-level claims we test and our alignment metrics and simulation design. Section 4 reports whether explanations align with the data and checks the result’s robustness; Section 5 isolates the drivers of misalignment. Section 6 develops the Rashomon-agreement diagnostic, and Section 7 concludes with guidance for practice.

2 Background and Related Work

SHAP and LIME. SHAP (Lundberg and Lee 2017) attributes a prediction to features using the Shapley value from cooperative game theory. Its appeal rests on two properties: *local accuracy* (feature contributions sum to the prediction minus its baseline) and *consistency* (making a feature more important cannot decrease its attribution). LIME (Ribeiro et al. 2016) perturbs an instance, weights the perturbations by proximity, and fits a sparse linear surrogate whose coefficients are read as local feature influences. Both are model-agnostic and report a direction (sign) and a strength (magnitude) per feature; global importance is usually taken to be the mean absolute attribution across instances.

Scrutiny in computer science. The ML literature has scrutinized both methods extensively, but at the *model* level—whether an explanation faithfully describes the fitted model: LIME is unstable across sampling and kernel choices (Garreau and von Luxburg 2020, Alvarez-Melis and Jaakkola 2018), explanations can be adversarially manipulated (Slack et al. 2020), and Shapley-value attributions have been criticized as ill-suited to human notions of importance (Kumar et al. 2020, Han et al. 2022). Notably, even the sharpest of these critiques concern specific failure contexts—models deliberately engineered to deceive the explainer (Slack et al. 2020), or counterfactual decision tasks the tools were not designed for (Fernández-Loría et al. 2022)—and none tests whether an explanation applied in good faith to a well-trained model recovers relationships in the *data*. Model-level faithfulness, even where it holds, does not imply data-level validity.

Engagement in business and management. The business and management community has engaged with these tools just as actively, but from an applied, consequential angle. One stream studies how explainable AI shapes managerial and user decision-making (Bauer et al. 2023, von Zahn et al. 2025), including in high-stakes domains such as healthcare (Hassan et al. 2025) and misinformation detection (Lee and Ram 2024), and the regulatory and compliance challenges it raises—especially in credit scoring (Mohammadi et al. 2025, Bückner et al. 2022, Chen et al. 2024).

A parallel methodological stream advances new or improved explanations, for instance contrasting feature attributions with counterfactuals (Fernández-Loría et al. 2022) or refining LIME’s sampling and surrogate-fitting stages (Kim et al. 2023). This body of work has meaningfully advanced how the field deploys and governs post hoc explainers.

Where our question sits. Crucially, both literatures evaluate explanations *of the model* ($\mathbf{X} \mapsto \hat{Y}$)—whether they faithfully or usefully describe the fitted predictor. Our question is one step further removed and, for empirical research, more consequential: whether explanations of $\mathbf{X} \mapsto \hat{Y}$ license conclusions about the data itself ($\mathbf{X} \mapsto Y$). This is the very inference that econometric tools—regression, causal inference, experiments—are built to support by making identification assumptions explicit, whereas post hoc explainers were designed only to describe a fitted model.

The Rashomon effect. For a given task there is often a *Rashomon set*: many models with nearly identical predictive performance but different internal structure and feature attributions (Semenova et al. 2022, Paes et al. 2023, Rudin et al. 2024). Each such model encodes a distinct hypothesis about $\mathbf{X} \mapsto Y$. Consequently, even a perfect explanation of a *single* model need not reflect the true relationship in the data—a point that motivates both our misalignment analysis and our diagnostic.

3 Measuring Alignment with the Data

Across our review, the interpretations researchers draw from SHAP and LIME reduce to two recurring *data-level* claims: *direction* claims (“ X_j has a positive/negative effect on Y ” when feature X_j receives a positive/negative attribution) and *strength* claims (“ X_1 matters more than X_2 ” when X_1 ’s aggregate importance is larger, often used to name the “most important drivers”). Both assume that a statement about the model ($\mathbf{X} \mapsto \hat{Y}$) transfers to the data ($\mathbf{X} \mapsto Y$). We now define two metrics that make these claims testable. The definitions are stated for a generic feature-attribution explainer $\mathcal{E}$, predictive model $\mathcal{M}$, and ground-truth data-generating process $\mathcal{G}$; SHAP and LIME arise as instances. Let $e_{\mathcal{E}}(\mathbf{x}^{(i)}, j)$ denote the attribution that $\mathcal{E}$ assigns to feature j at instance i .

3.1 Direction Alignment

Direction alignment asks, *for each feature*, whether the direction of change implied by the explainer matches the true direction of change in the data. We compare responses to single-feature perturbations that hold other features fixed. For instance $\mathbf{x}^{(i)}$, feature j , and magnitude δ , let $\mathbf{x}_{j,\delta}^{(i)} = (x_1^{(i)}, \dots, x_j^{(i)} + \delta, \dots, x_d^{(i)})$, with δ drawn from a method-independent grid $\Delta = \{m \cdot d : m \in \{-M, \dots, -1, 1, \dots, M\}\}$. The *explainer-indicated* and *true* directional responses are

$$\Delta e_j^{(i)}(\delta) = e_{\mathcal{E}}(\mathbf{x}_{j,\delta}^{(i)}, j) - e_{\mathcal{E}}(\mathbf{x}^{(i)}, j), \quad \Delta Y_j^{(i)}(\delta) = \mathcal{G}(\mathbf{x}_{j,\delta}^{(i)}) - \mathcal{G}(\mathbf{x}^{(i)}). \quad (1)$$

Requiring strict sign agreement is unreasonable when the true effect is negligible, so we treat small true changes as directionally uninformative:

$$I_{ij}^{\text{dir}}(\delta) \triangleq \begin{cases} \mathbb{1}(\Delta e_j^{(i)}(\delta) \Delta Y_j^{(i)}(\delta) > 0) & \text{if } |\Delta Y_j^{(i)}(\delta)| > \epsilon_1, \\ \mathbb{1}(|\Delta e_j^{(i)}(\delta)| < \epsilon_2) & \text{otherwise.} \end{cases} \quad (2)$$

Table 1: Data-generation factors. Four varied factors at three levels each yield $3^4 = 81$ datasets.

Factor	Variable	Type	Values
Sample size	Num_instance	Fixed	5,000
Noise std. dev.	Noise_std	Fixed	0.1
Number of features	Num_feature	Varied	10, 20, 30
Correlation strength	Corr_strength	Varied	0, 0.5, 1
Nonlinear terms	Nonlinear_num	Varied	0, 5, 10
Interaction terms	Interaction_num	Varied	0, 5, 10

Aggregating over instances S and perturbations Δ , the direction-alignment score for feature j is

$$\rho_{\text{dir},j} = \frac{1}{|S| |\Delta|} \sum_{i=1}^{|S|} \sum_{\delta \in \Delta} I_{ij}^{\text{dir}}(\delta) \in [0, 1]. \quad (3)$$

For SHAP, $e_{\mathcal{E}}(\mathbf{x}^{(i)}, j)$ is the Shapley value and $\Delta e_j^{(i)}$ is evaluated directly on the perturbation grid. For LIME, the local coefficient $w_j^{(i)}$ gives $\Delta e_j^{(i)}(\delta) = w_j^{(i)} \cdot \delta$ for small δ (we use $\delta = 0.1 \text{ Std}$), keeping the evaluation within the surrogate’s local neighborhood.

3.2 Strength Alignment

Strength alignment asks whether the explainer recovers the *relative* importance of features. Let $I_{\mathcal{E}}(j)$ and $I_{\mathcal{G}}(j)$ be the global importance of feature j under the explainer and the ground truth, with induced rankings $R_{\mathcal{E}}$ and $R_{\mathcal{G}}$. We define strength alignment as the Spearman rank correlation

$$\rho_{\text{strength}} = 1 - \frac{6 \sum_{j=1}^d (R_{\mathcal{E}}(j) - R_{\mathcal{G}}(j))^2}{d(d^2 - 1)} \in [-1, 1]. \quad (4)$$

Global importance is the mean absolute attribution across instances; the ground-truth importance $I_{\mathcal{G}}(j)$ is obtained by applying the same explainer to $\mathcal{G}$. Values near 1 indicate faithful recovery of the importance ranking; values near or below 0 indicate weak or systematically incorrect ranking.

3.3 Simulation Design

The first step of our evaluation is to construct data in which the true $\mathbf{X} \mapsto Y$ is known and can serve as the benchmark: we simulate each dataset by drawing features for 5,000 instances and applying a ground-truth process $\mathcal{G}$ that computes a weighted sum of untransformed, squared, and pairwise-interaction terms (with random coefficients) and passes it through a sigmoid to generate binary outcomes. We vary four factors—number of features, correlation strength, nonlinear terms, and interaction terms—each at three levels, holding sample size and noise fixed, for a total of **81 datasets** (Table 1). For each dataset we follow a standard pipeline: train a predictive model $\mathcal{M}$ on 80% of the data (searching over gradient-boosted trees, random forests, MLPs, and SVMs with cross-validated tuning) and evaluate on the remaining 20%. We then apply SHAP and LIME to $\mathcal{M}$ and compute the alignment metrics against $\mathcal{G}$.

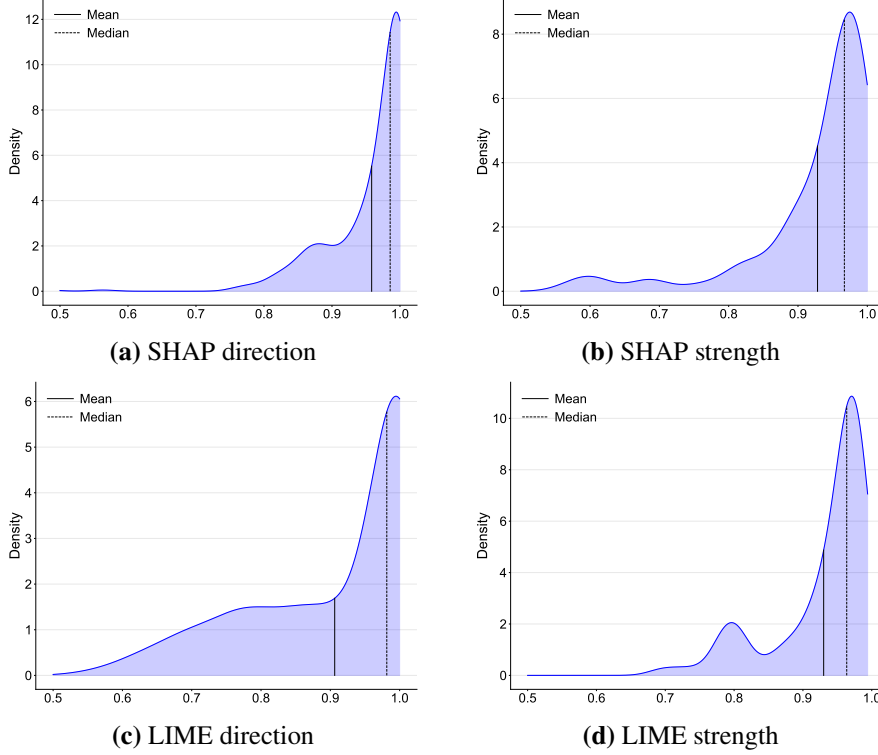


Figure 2: Distributions of direction- and strength-alignment scores for the best model of each of the 81 datasets. Average alignment is high, but every distribution has a pronounced long left tail.

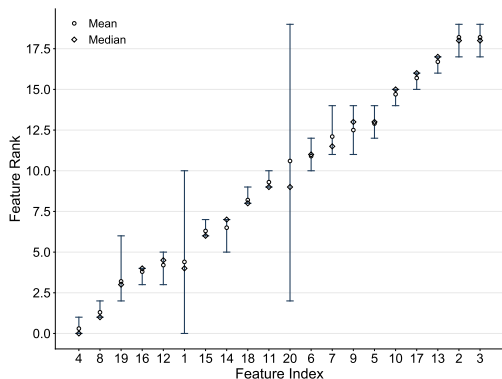
4 Do Post Hoc Explanations Align with the Data?

For each dataset we summarize alignment by averaging over the five features with the largest mean attribution, then examine the distribution of these scores across all 81 datasets. Figure 2 shows the results, which yield two findings.

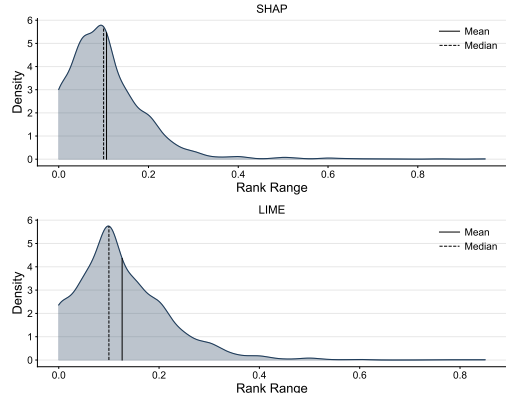
Finding 1 (high average performance). On average, both explainers align well with $\mathcal{G}$ in direction and strength, and SHAP consistently recovers feature directions better than LIME.

Finding 2 (long left tails undermine dataset-level reliability). Every distribution has a pronounced long left tail: although the average is high, a nontrivial subset of datasets aligns poorly. For SHAP the heavy tail is in *strength*, falling to ≈ 0.5 (severe mis-ranking) for some datasets; for LIME it is in *direction*, also falling to ≈ 0.5 (near-random sign agreement). Because a researcher usually lacks an ex ante indicator of when an explainer is in this low-alignment regime, high average alignment masks real dataset-level risk when drawing conclusions about $\mathbf{X} \mapsto Y$.

Robustness checks. We probed whether this misalignment result depends on specific implementation or evaluation choices with three checks, each motivated by a natural stabilizing intuition. (1) *Averaging over a Rashomon set:* averaging attributions across the ten best models improves direction recovery for SHAP, but the alignment distributions retain their pronounced left tails—severe failures in recovering direction and strength still occur with nontrivial frequency. (2) *Explainer hyperparameters:* varying SHAP’s sampling budget and LIME’s kernel bandwidth and neighborhood size, and averaging LIME across random seeds, yield small gains but do not overturn the pattern. (3) *Metric parameters:* varying the perturbation magnitude δ does not materially alter the shape of the alignment distributions. Across all settings, data misalignment persists even under configurations designed to be favorable to alignment, such as consensus explanations averaged across



(a) Range of SHAP ranks across a Rashomon set (one dataset)



(b) Distribution of rank ranges (all datasets)

Figure 3: Evidence of the Rashomon effect on post hoc explanations: equally accurate models disagree substantially on feature importance.

equally accurate models or explainers averaged over random seeds at tuned hyperparameters.

5 What Drives Misalignment?

We examine three classes of drivers: the predictive performance of $\mathcal{M}$, model multiplicity (the Rashomon effect), and intrinsic data properties.

Predictive performance is necessary. Explainers describe $\mathcal{M}$, not $\mathcal{G}$; if $\mathcal{M}$ poorly approximates $\mathcal{G}$, no explainer can align with the truth. Constructing models across accuracy buckets $[0.70, 0.75), \dots, [0.85, 0.90)$ for each dataset and regressing alignment on the ordered bucket, we find accuracy has a statistically significant effect, particularly for SHAP direction: a 0.05 increase in test accuracy is associated with an average increase of 0.104 in SHAP’s direction alignment. Predictive accuracy is therefore a genuine *precondition* for alignment.

The Rashomon effect makes accuracy insufficient. Accuracy alone cannot pin down $\mathcal{G}$, because a large Rashomon set contains many equally accurate models that map $\mathbf{X} \mapsto \hat{Y}$ through different internal logic. To illustrate, for each dataset we took the 10 best models (test accuracies within 3% of one another), applied SHAP, and ranked features by mean absolute Shapley value. Figure 3(a) shows, for a representative dataset, the range of ranks each feature receives across the 10 models: some feature ranks are stable, but others swing wildly—one feature is ranked both most important (rank 1) and largely unimportant (rank 11 of 20) by different, equally accurate models. Aggregating across all datasets (Figure 3(b), normalized by feature count), ranks vary by about 10% on average with a long right tail. High accuracy thus cannot guarantee alignment: even in best-case predictive settings, the Rashomon effect means many distinct decision rules fit the observed data equally well, and accuracy alone cannot identify which of them reflects the true $\mathbf{X} \mapsto Y$ relationship. The asymmetry is important—low accuracy is strong evidence *against* alignment, but high accuracy is no guarantee of it.

Feature correlation is the dominant data-level driver. To quantify how data properties shape alignment, we regress each alignment metric on the data-generation factors; Figure 4 reports the OLS coefficients. Correlation strength, the number of nonlinear terms, and the number of

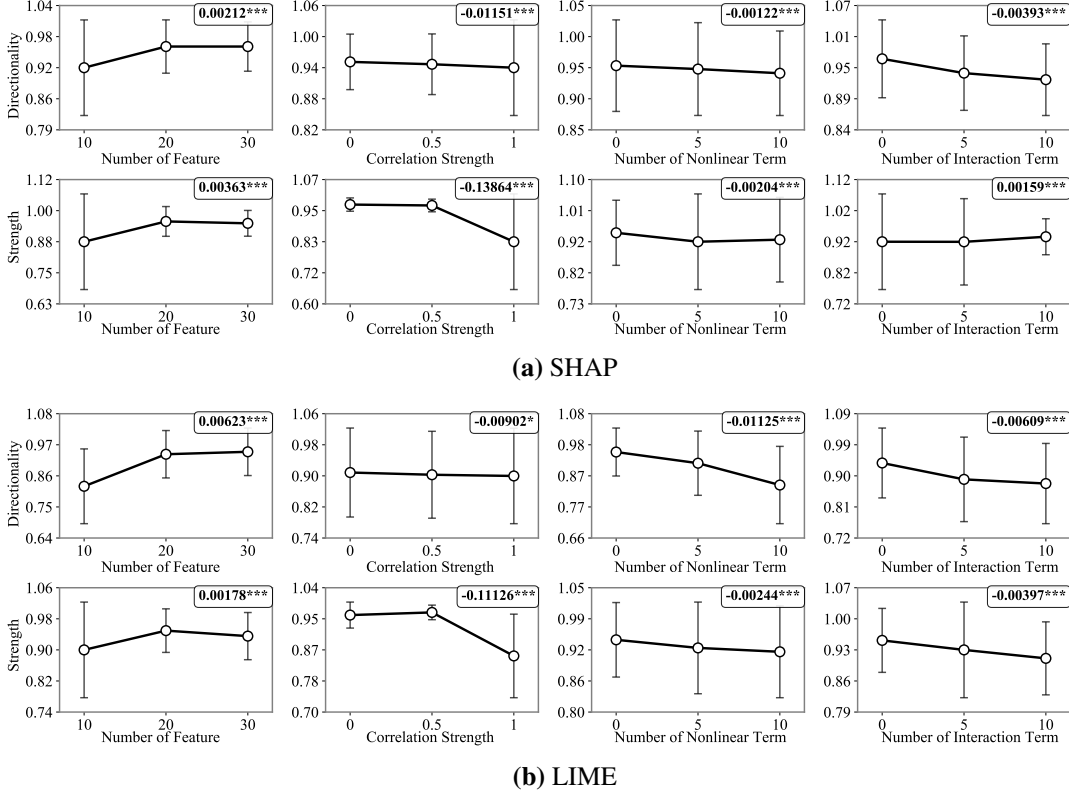


Figure 4: Impact of data characteristics on alignment. Numbers at the top right of each panel are univariate OLS coefficients. Correlation strength is the dominant driver of misalignment.

interaction terms are all significantly associated with reduced alignment, but correlation strength has by far the largest and most consistent negative effect, especially on strength alignment. When features are correlated, several variables can substitute for one another, enlarging the set of near-optimal models with different internal attributions and thereby increasing explanation disagreement. Nonlinearity and interactions act through the same mechanism—more admissible functional forms, less identifiability—while the number of features has a negligible ($< 1\%$) effect.

6 Diagnosing Reliability via Rashomon Agreement

Since explanations are reliable on average but fail on a nontrivial subset of datasets, a practical question is: *when can a given explanation be trusted as evidence about the true $\mathbf{X} \mapsto Y$ relationship?* We propose that disagreement among equally accurate models—which we call *Rashomon agreement*—is an observable signal. If models in a Rashomon set disagree, many distinct hypotheses about $\mathbf{X} \mapsto Y$ fit the data equally well, so any single model’s explanation is unlikely to match $\mathcal{G}$. High agreement indicates a more constrained hypothesis space, increasing confidence that a single model’s explanation reflects data-supported structure rather than a model-specific artifact.

Agreement cannot be observed directly, so we measure it two ways. *Prediction agreement* is the fraction of instances on which two models produce the same predicted label. *Explanation agreement* is the Spearman correlation between two models’ feature-importance rankings (mean absolute attribution), a scale-invariant comparison of *how* each model attributes importance. Each is computed pairwise within a Rashomon set and averaged to a dataset-level score.

Table 2: Correlation between Rashomon agreement and explanation alignment. Explanation-based agreement is far more informative than prediction-based agreement, especially for strength.

Explainer	Agreement type	Direction alignment	Strength alignment
SHAP	Explanation agreement	0.326	0.792
	Prediction agreement	0.344	0.088
LIME	Explanation agreement	0.217	0.695
	Prediction agreement	0.131	0.023

Table 2 reports how each agreement metric correlates with alignment. Two results stand out. First, *explanation* agreement is strongly correlated with alignment—up to 0.792 for SHAP and 0.695 for LIME on strength—so it is a usable signal of reliability. Second, *prediction* agreement is far weaker (near zero for strength): models can agree on outputs while relying on entirely different feature attributions. It is disagreement in attributions, not in predictions, that flags heightened risk of misinterpreting $\mathbf{X} \mapsto Y$. A secondary asymmetry is that SHAP alignment is more predictable from agreement than LIME’s, whose alignment remains noisier even when models agree.

Examining which data properties drive agreement clarifies the picture. Prediction agreement falls sharply with data complexity (e.g., an added interaction term is associated with a 16.9% decrease), whereas explanation agreement is less tied to complexity—only correlation strength has a significant, small effect. This reframes explanation reliability as a *dataset-level* property: low agreement, especially low explanation agreement, is an actionable warning that many equally accurate models tell different stories. Practically, a researcher can run the pipeline across several equally accurate models and data splits and check whether the explanations agree before drawing any $\mathbf{X} \mapsto Y$ conclusion.

7 Conclusion

Post hoc explainers are increasingly used in business research not only to explain models but to make claims about data. We document that this practice is widespread (about 42.5% of 181 reviewed papers) and show, on simulated data with known ground truth, that it is unreliable: SHAP and LIME recover feature direction and strength well on average but fail on a nontrivial fraction of datasets, even when predictive accuracy is high. These failures are structural—the Rashomon effect means many equally accurate models encode different attributions, and feature correlation amplifies the ambiguity. Disagreement among such models is a powerful diagnostic, whereas common robustness strategies offer only limited mitigation.

The implication is not that post hoc explainers are worthless, but that their role must be reframed. They are well suited to **proposing** hypotheses—surfacing candidate variables, mechanisms, or heterogeneity worth investigating—but not to **validating** them. Any hypothesis an explainer generates should be treated as provisional and confirmed with methods built for identification: regression-based approaches, causal inference, or experiments. Several studies already model this workflow, using machine learning to surface candidate drivers and then turning to causal forests (Zhang and Luo 2023), controlled experiments (Feng et al. 2025), or field experiments (Senoner et al. 2022) to validate them. Keeping hypothesis generation and hypothesis validation distinct lets post hoc explainers complement the empirical toolkit without overstating their inferential content.

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